

1. Problem Overview

This project focuses on predicting spot truckload freight rates (posted_rate in USD) from load characteristics. The goal is to build an accurate, reproducible pricing model to predict rates for 12,000 unlabeled validation shipments and generate a 31-day rate forecast for a fixed benchmark corridor (Lexington to Fort Wayne) in December 2025.

2. Dataset Summary

- **Historical Data:** 48,000 labeled loads (2025-01-01 to 2025-10-31) across 14 columns with ground-truth posted_rate.
- **Validation Data:** 12,000 unlabeled loads spanning Nov 1 to Dec 31, 2025 (load_id TE-000001 to TE-012000).
- **December Test Set:** 31 daily scenarios for fixed corridor Lexington -> Fort Wayne (360 miles, Dry Van, 32,000 lb).

3. Exploratory Data Analysis & Findings

- **Mileage Impact:** Distance is the strongest driver of freight rate ($r \approx 0.91$). Transformed via $\log_{1p}(\text{distance})$ to linearize.
- **Target Skewness:** posted_rate ranges from USD 57 to USD 25,533 with heavy right-skew (~ 1.90), motivating log-scale training.
- **Data Quality:** Identified 292 negative weight records (sign-entry errors) fixed via $\text{abs}(\text{weight})$. Missing weights (0.62%) and market_index (0.78%) were preserved for training-fold median imputation. No duplicate loads found.

4. Data Cleaning

- **Weight Correction:** Negative weights were converted to positive values via $\text{abs}(\text{weight})$; zero training records were dropped.
- **Missing Values:** Missing numerical features were left as NaN and handled via median imputation inside the pipeline.
- **Date Parsing:** Standardized all date strings to datetime format; raw dataset files remain immutable.

5. Feature Engineering

- **Coordinates:** Standard-scaled pickup and delivery latitude and longitude.
- **Distance:** $\log_{1p}(\text{distance})$ followed by standard scaling to improve linear model and tree split efficiency.
- **Categoricals:** One-hot encoding for equipment type, pickup city, and delivery city with unknown handling.
- **Lane Frequency:** Combined pickup and delivery (pickup -> delivery) and frequency-encoded using training fold counts.
- **Calendar Features:** Sine and cosine transformations for month ($T=12$), quarter ($T=4$), and day-of-week ($T=7$).
- **Exclusions:** load_id, market_index, and quote_signal were excluded from features to maintain pipeline simplicity.

6. Preventing Data Leakage

- Time Split Priority:** The dataset was partitioned by date before fitting any scaler, encoder, or imputer.
- Encoding Safety:** Lane frequencies were computed solely on training rows; unseen test lanes default to training minimum.
- Target Isolation:** Target was modeled as $\log_{1p}(\text{posted_rate})$; no target-derived statistics were used as input features.
- Fixed Pipeline:** Validation and December datasets were transformed using the pre-fitted pipeline without refitting.

7. Validation Strategy

- Time-Based Split:** Cutoff date set at 2025-08-31 (80% chronological split), simulating realistic forward deployment.
- Split Sizes:** Training fold: 38,477 rows (Jan–Aug 2025); Internal validation fold: 9,523 rows (Sep–Oct 2025).
- Evaluation Metric:** Evaluated on original USD dollar scale after reversing the log transform with $\exp m1$.

8. Model Comparison & Benchmarks

Six models were evaluated on the internal validation set under identical preprocessing:

Model Architecture	Holdout MAE (USD)	Holdout RMSE (USD)	Holdout R ²	Notes
CatBoost (Tuned)	133.0854	641.1042	0.8235	Best performance (lowest MAE & RMSE)
CatBoost (Baseline)	143.1829	644.3404	0.8217	Default hyperparameters
LightGBM	145.9222	644.9609	0.8214	Fast training, competitive boosting
Linear Regression	162.1602	650.9376	0.8181	Simple baseline with higher linear error
Random Forest	167.6709	662.6681	0.8114	Higher variance across trees
XGBoost	167.6918	660.4121	0.8127	Higher error on distribution tails

9. Final Model & Hyperparameters

- Chosen Estimator:** CatBoostRegressor tuned via TimeSeriesSplit grid search on the training fold.

- depth: 6
 - iterations: 500
 - l2_leaf_reg: 1
 - learning_rate: 0.03
 - loss_function: 'RMSE'
 - random_state: 42

- Model Export:** Model saved to outputs/catboost_model.cbm and full preprocessing pipeline to outputs/pipeline.joblib.

10. December 2025 Forecast Analysis

- **Benchmark Route:** Lexington -> Fort Wayne (360 miles, Dry Van, 32,000 lb). City coordinates mapped from training data.
- **Pattern Rationale:** With route, load size, month (12), and quarter (4) fixed, the daily variation across December smoothly captures the learned day-of-week rate cycle (oscillating between USD 769.92 and USD 787.05).

Candidate: December 2025 Predicted Load Rate

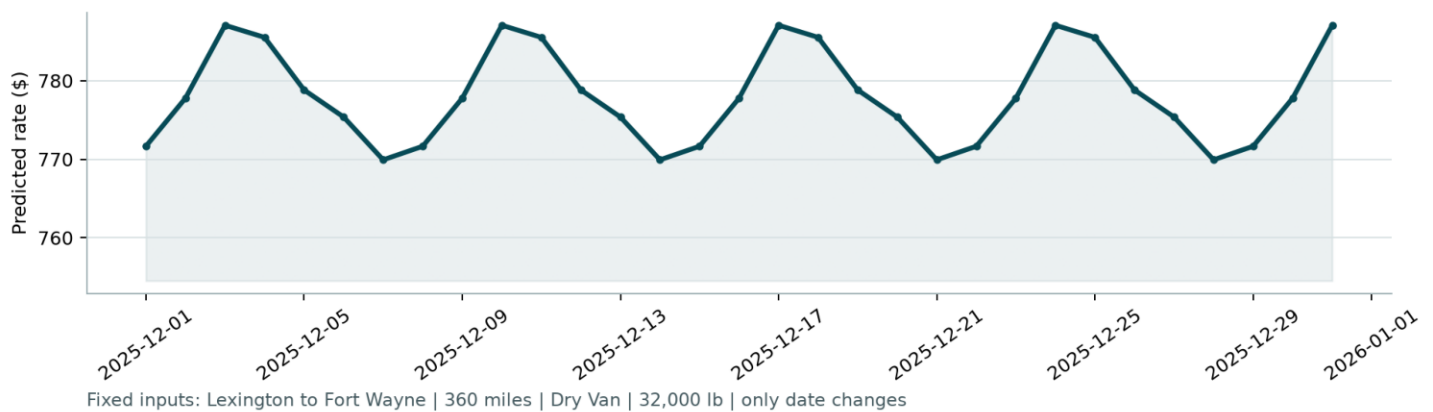


Figure 1: Generated December 2025 rate trajectory for Lexington -> Fort Wayne test corridor.

11. Limitations & Considerations

- **Validation Scope:** Reported metrics reflect the Sep-Oct 2025 holdout; final validation score is computed on unseen data.
- **Routing Distance:** Minor geodesic discrepancies between raw lat/lon coordinates and stated distance were kept as given.
- **Feature Selection:** market_index and quote_signal were excluded from the final model to keep the pipeline clean and stable.

12. Conclusion

- **Pipeline Summary:** The final CatBoost pipeline is modular, lightweight, and completely reproducible without data leakage.
- **Output Verification:** 12,000 validation predictions and 31 December forecasts satisfy all format, ID, and positivity checks.
- **Scorer Validation:** The official scoring script passes cleanly with exit code 0 and verifies all prediction files.